

# Data Science Applications to Politics Research

## Week 2

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GV330

## 1 Modes of Statistical Inference

- Descriptive Inference
- Predictive Inference
- Causal Inference

## 2 “Big data” and Causal Inference

## 3 Credibility Crisis in the Social Sciences

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# Three Modes of Statistical Inference

1. Descriptive Inference: summarizing and exploring data
  - Inferring “ideal points” from roll-call votes
  - Inferring “topics” from texts and speeches
  - Inferring “economic development” from satellite images
2. Predictive Inference: forecasting out-of-sample data points
  - Inferring future state failures from past failures
  - Inferring election outcomes from polls
  - Inferring individual level behavior from aggregate data
3. Causal Inference: predicting *counterfactuals*
  - Inferring the effect of a policy on voter turnout
  - Inferring the impact of social media use on political polarization
  - Inferring whether severe weather events make people more likely to believe in climate change

# Descriptive Inference Example:

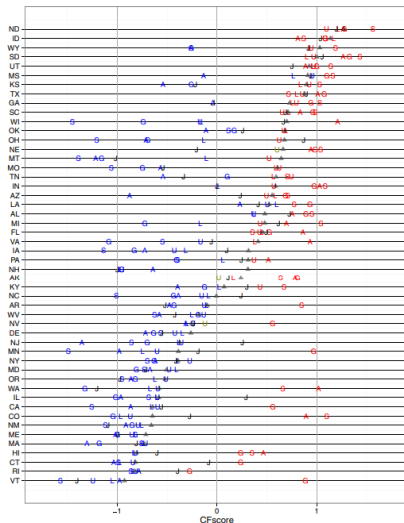
## Mapping the Ideological Marketplace (Bonica 2014)

Develops method of measuring ideology of political candidates and campaign contributors using campaign finance data

- Data: Over 100 million campaign finance records from U.S. state and federal elections
- Generates **ideal points** using common pool of contributors who give to candidates across institutions and levels of politics
  - common-space campaign finance scores (CFscores)
- Assumption: contributors will on average prefer **ideologically proximate** candidates to those who are more distant.
  - Contribution amounts allocated based on distance between a candidate's ideal point and the contributor's ideal point
- Relies on donors who give to candidates for a variety of offices to bridge across institutions and levels of politics

Bonica, A. (2014). Mapping the ideological marketplace. *American Journal of Political Science*, 58(2), 367-386.

FIGURE 3 Ideological Summary of State Politics (2010)



Note: The symbols are interpreted as follows: G = Governor, A = Attorney General, S = Secretary of State, J = State Supreme Court (median), L = Lower Legislative Chamber (median), U = Upper Legislative Chamber (median), black triangle = mean ideal point of candidates elected in the state. The symbols are color coded by party (Dem = Blue; Rep = Red).

# Predictive Inference Example:

## Voting behaviour (Nickerson & Rogers 2014)

- Campaigns use data to construct **predictive models** for more efficient targeting
- These models produce three types of “**predictive scores**” for each citizen in the voter database:
  - **behavior scores**: use past behavior and demographic information to calculate probabilities that citizens turn out, donate, etc.
  - **support scores**: survey a sample of citizens about their candidate/issue support. Use to gauge aggregate preferences.
  - **responsiveness scores**: predict how citizens will respond to campaign outreach using results of randomized field experiments. Model heterogeneous treatment effects and use to predict treatment responsiveness in target population.
    - Note that this can be considered *causal inference* – see next section

# Example: Predicting voting behaviour (Nickerson & Rogers 2014)

## “Big data”

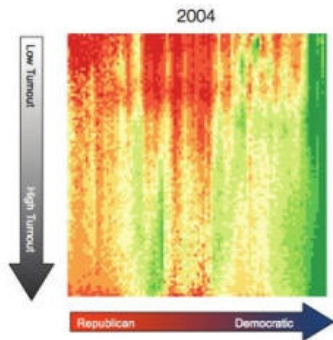
- voter files
- consumer and property data
- Census data (neighborhood characteristics)
- previous engagement (e.g., donating, volunteering)

## Analyses

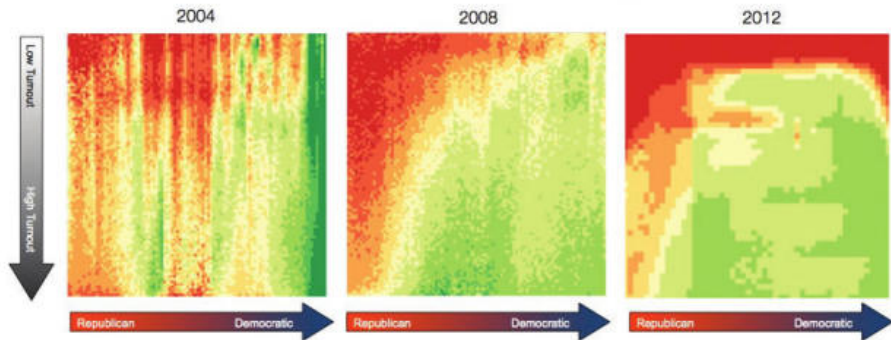
- Simple regression techniques (e.g., OLS)
- Supervised learning algorithms



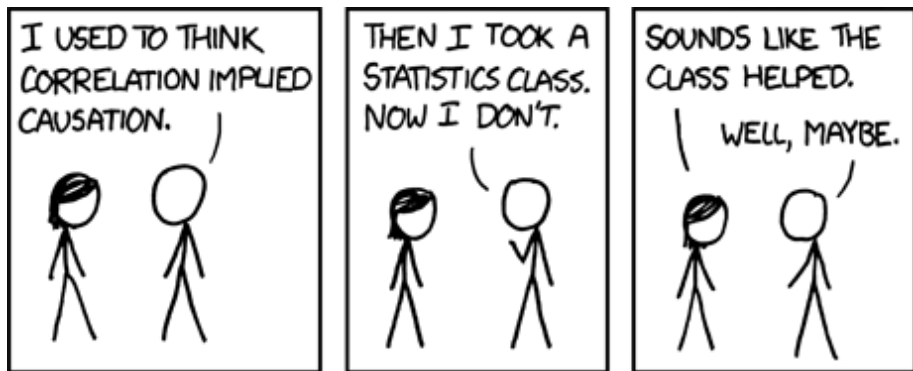
Ohio Contacts Over Three Presidential Cycles



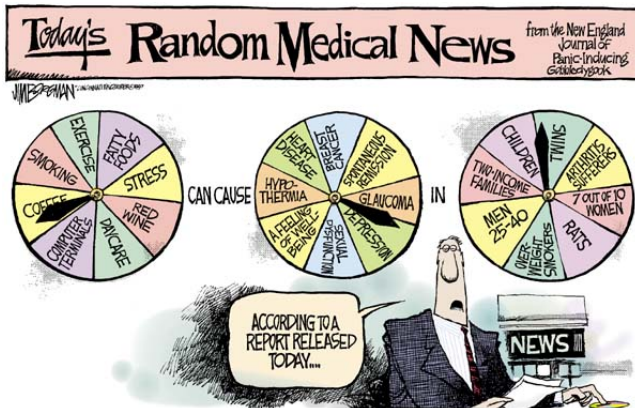
Ohio Contacts Over Three Presidential Cycles



# Causal Inference



# Experiments vs. Observational Studies



Lancet 2003: comparing among individuals with the same age, gender, blood pressure, diabetes, and smoking, those with higher vitamin C levels have lower levels of obesity, lower levels of alcohol consumption, are less likely to grow up in working class, etc.

# What is Causal Inference?

Causal inference = inference about **counterfactuals**

A **causal effect** is a change in some feature of the world  $Y$  that *would result* from a change to some other feature of the world  $D$ .

Examples:

- Incumbency advantage:  
What *would* have been the election outcome if the candidate had not been an incumbent?
- Resource curse:  
What *would* have been the GDP growth rate without oil?
- Democratic peace:  
*Would* the two countries have fought each other if they had been both autocratic?
- Policy intervention:  
How many more disadvantaged youths *would* get employed under the new job training program?

# Fundamental problem of causal inference



# Causal Inference Example: GOTV Experiment (Gerber et al 2008)

- Voter turnout theories based on rational self-interested behavior generally fail to predict significant turnout unless they account for the utility that citizens receive from performing their civic duty.
- Two aspects of this type of utility, intrinsic satisfaction from behaving in accordance with a norm and extrinsic incentives to comply
- Gerber, Green, and Larimer (2008) test intrinsic motives in a large scale field experiment by applying varying degrees of extrinsic pressure on voters using a series of mailings to 180,002 households before the August 2006 primary election in Michigan.
  - $Y_i$ : voted in primary (yes/no)
  - $D_i$ : type of mailing

# Example: GOTV Experiment (Gerber et al 2008)

- **Civic Duty:**
  - Encouraged to vote
- **Hawthorne:**
  - Encouraged to vote
  - Told that researchers would be checking on whether they voted
- **Self:**
  - Encouraged to vote
  - Told that whether one votes is a matter of public record
  - Shown whether members of their own household voted in the last two elections
- **Neighbors:**
  - Like **Self** but in addition recipients are shown whether the neighbors on the block voted in the last two elections elections



# Social Pressure Treatment (Gerber et al 2008)

Dear Registered Voter:

## WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

## DO YOUR CIVIC DUTY — VOTE!

---

| MAPLE DR                   | Aug 04 | Nov 04 | Aug 06 |
|----------------------------|--------|--------|--------|
| 9995 JOSEPH JAMES SMITH    | Voted  | Voted  | _____  |
| 9995 JENNIFER KAY SMITH    |        | Voted  | _____  |
| 9997 RICHARD B JACKSON     |        | Voted  | _____  |
| 9999 KATHY MARIE JACKSON   |        | Voted  | _____  |
| 9999 BRIAN JOSEPH JACKSON  |        | Voted  | _____  |
| 9991 JENNIFER KAY THOMPSON |        | Voted  | _____  |
| 9991 BOB D. THOMPSON       |        | Voted  | _____  |

# Experimental results (Gerber et al 2008)

|                         | <b>Control</b><br>(Not Mailed) | <b>Civic Duty</b><br>(Encouraged to vote) | <b>Hawthorne</b><br>(Encouraged & Monitored) | <b>Self</b><br>(Encouraged, Monitored, Shown Own Past Voting) | <b>Neighbors</b><br>(Encouraged, Monitored, Shown Own & Others' Past Voting) |
|-------------------------|--------------------------------|-------------------------------------------|----------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Percent Voting</b>   | <b>29.7%</b>                   | <b>31.5%</b>                              | <b>32.2%</b>                                 | <b>34.5%</b>                                                  | <b>37.8%</b>                                                                 |
| <b>N of Individuals</b> | <b>191,243</b>                 | <b>38,218</b>                             | <b>38,204</b>                                | <b>38,218</b>                                                 | <b>38,201</b>                                                                |

# Covariate balance (Gerber et al 2008)

With  $n \simeq 180,000$ , covariates are almost perfectly balanced:

**TABLE 1. Relationship between Treatment Group Assignment and Covariates (Household-Level Data)**

|                | Control | Civic Duty | Hawthorne | Self   | Neighbors |
|----------------|---------|------------|-----------|--------|-----------|
|                | Mean    | Mean       | Mean      | Mean   | Mean      |
| Household size | 1.91    | 1.91       | 1.91      | 1.91   | 1.91      |
| Nov 2002       | .83     | .84        | .84       | .84    | .84       |
| Nov 2000       | .87     | .87        | .87       | .86    | .87       |
| Aug 2004       | .42     | .42        | .42       | .42    | .42       |
| Aug 2002       | .41     | .41        | .41       | .41    | .41       |
| Aug 2000       | .26     | .27        | .26       | .26    | .26       |
| Female         | .50     | .50        | .50       | .50    | .50       |
| Age (in years) | 51.98   | 51.85      | 51.87     | 51.91  | 52.01     |
| N =            | 99,999  | 20,001     | 20,002    | 20,000 | 20,000    |

*Note:* Only registered voters who voted in November 2004 were selected for our sample. Although not included in the table, there were no significant differences between treatment group assignment and covariates measuring race and ethnicity.

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# “Big data” and Causal Inference

Having a lot of data can help us make causal inferences, under certain conditions. For example:

- If we have a randomized experiment with many participants
- In a regression discontinuity design (RDD) context, it helps to have many observations around the threshold

“Big data” and research design for causal inference are not substitutes.

- Need **exogenous variation** and a solid research design in order to isolate causal effects (“identification precedes estimation”)
- Credibly identifying a source of exogenous variation requires **creative insight** about, and **substantive knowledge** of, the phenomenon under study.
- Computational techniques cannot replace “**actual people doing actual thinking about social phenomena.**” (Ashworth, Berry, & Bueno de Mesquita 2015)

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# What is the problem?

Adapted from: Clark, J., Desposato, S., and McIntosh, C. 2017. "How to improve the credibility of (your) social science: A practical guide for researchers." Policy Design and Evaluation Lab (PDEL). University of California, San Diego.

JELLY BEANS  
CAUSE ACNE!

SCIENTISTS!  
INVESTIGATE!

BUT WE'RE  
PLAYING  
MINECRAFT!  
... FINE.



WE FOUND NO  
LINK BETWEEN  
JELLY BEANS AND  
ACNE ( $P > 0.05$ ).



THAT SETTLES THAT.

I HEAR IT'S ONLY  
A CERTAIN COLOR  
THAT CAUSES IT.

SCIENTISTS!

BUT  
MINECRAFT!





WE FOUND NO  
LINK BETWEEN  
PURPLE JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
BROWN JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
PINK JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
BLUE JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
TEAL JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
SALMON JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
RED JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
TURQUOISE JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
MAGENTA JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
YELLOW JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
GREY JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
TAN JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
CYAN JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND A  
LINK BETWEEN  
GREEN JELLY  
BEANS AND ACNE  
( $P < 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
MAUVE JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
BEIGE JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
LILAC JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
BLACK JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).

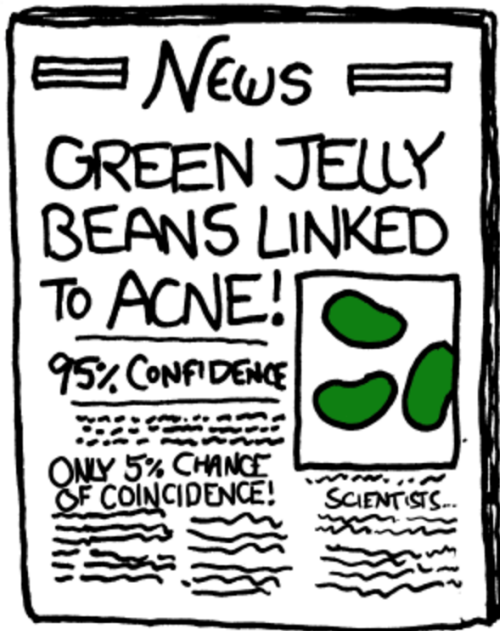


WE FOUND NO  
LINK BETWEEN  
PEACH JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).



WE FOUND NO  
LINK BETWEEN  
ORANGE JELLY  
BEANS AND ACNE  
( $P > 0.05$ ).





Source: XKCD

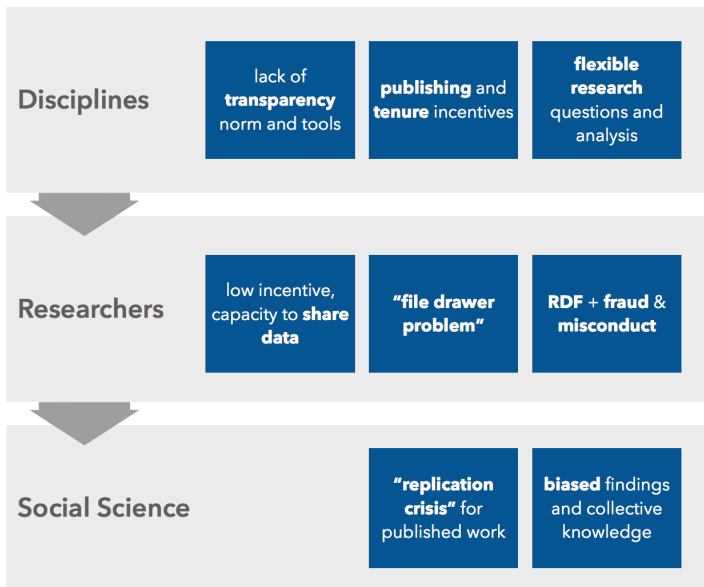
# We often hear about ...

We often hear about ...

- 1 **“Replication crisis”**—studies fail to replicate (psych, econ, polisci, medicine, etc.)
- 2 **Publication bias**—published studies only represent fraction of results, biased toward statistically significant findings
- 3 **P-hacking/researcher degrees of freedom**—published studies use only a fraction of possible specifications, biased toward significance
- 4 **Misconduct/fraud**

→ adds up to **biased body of knowledge**

# Why do we have this credibility crisis?



# “Replication Crisis”

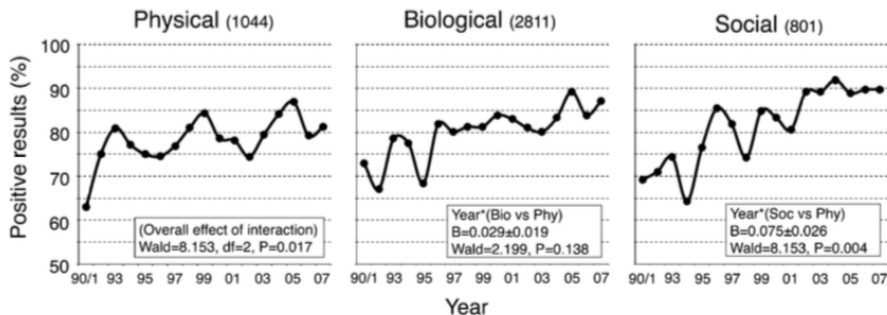
Social, behavioral, and medical studies often don't replicate.

- **Ideally**, replications determine if original results are robust or due to *random chance*.
- **In reality**, failure to replicate often a result of ...
  - Lack of transparency in sharing data/code
  - Errors in data/code
  - Misconduct or fraud

# Publication Bias (aka the “file drawer problem”)

- **Problem:** Studies more likely to be submitted/published when findings are significant → studies with null findings are hidden
- **Result:** Biased evidence base: we're missing full universe of studies and results; what gets published could be due to random chance (e.g., if we expect 5% of results of all studies to be significant)

Increase in % of papers with positive (i.e., they support the tested hypothesis) results over time, across scientific disciplines:





In a **known population** of conducted studies: strong results 60pp more likely to be written up than null results, 40pp more likely to be published:

**Table 3. Cross-tabulation between statistical results of TESS studies and their publication status (column percentages reported).** Pearson  $\chi^2$  test of independence:  $\chi^2(6) = 80.3, P < 0.001$ .

|                           | Null (%) | Mixed (%) | Strong (%) |
|---------------------------|----------|-----------|------------|
| Not written               | 64.6     | 12.2      | 4.4        |
| Written but not published | 14.6     | 39.0      | 34.1       |
| Published (non-top-tier)  | 10.4     | 37.8      | 38.4       |
| Published (top-tier)      | 10.4     | 11.0      | 23.1       |
| Total                     | 100.0    | 100.0     | 100.0      |